

Chapter 5 was amended by [Local Law 77 of 2023](#). This law has an effective date of June 10, 2023.

CHAPTER 5

GENERAL BUILDING HEIGHTS AND AREAS; SEPARATION OF OCCUPANCIES

SECTION BC 501 GENERAL

501.1 Scope. The provisions of this chapter control the height and area of structures hereafter erected and additions to existing structures, including separation of occupancies.

501.2 Address identification. Buildings shall be provided with address numbers or letters in accordance with Section 3-505 of the *Administrative Code* and the rules issued thereunder by the borough president of the borough in which the property is located. Except as otherwise provided by the rules of the borough president in which the property is located, characters shall be a minimum 4 inches (102 mm) high and a minimum of 0.5 inch (12.7 mm) wide. They shall be installed on a contrasting background and be plainly visible from the street or road fronting the property. Where access is by means of a private road and the building address cannot be viewed from the public way, an appropriately placed monument, pole or sign or any other means authorized by the borough president of the borough in which the property is located shall be used to identify the structure.

501.3 Fire Department access. Access shall be provided to the Fire Department in accordance with Sections 501.3.1 and 501.3.2.

501.3.1 Frontage. Every building, exclusive of accessory buildings, shall have at least 8 percent of the total perimeter of the building adjoining a street or frontage space. For the purposes of this section, building perimeter shall be measured at that story having the maximum enclosed floor area; and buildings provided with a front yard or front setback no deeper than 30 feet (9144 mm) in compliance with the *New York City Zoning Resolution* shall be considered as adjoining the street or frontage space.

501.3.2 Building access. Provisions shall be made for access by the Fire Department to every building in accordance with the *New York City Fire Code* and Section 903.2.11.1 of this code. In addition, multiple dwellings shall comply with Section 54 of the *New York State Multiple Dwelling Law*, as applicable.

SECTION BC 502 DEFINITIONS

502.1 Definitions. The following terms are defined in Chapter 2:

AREA, BUILDING.

BASEMENT.

CELLAR.

COMBINED HEAT AND POWER SYSTEMS.

EQUIPMENT PLATFORM.

FIRE LANE.

FRONTAGE SPACE.

GRADE PLANE.

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HEIGHT, BUILDING.

MEZZANINE.

STORY.

STORY ABOVE GRADE PLANE.

SECTION BC 503 GENERAL BUILDING HEIGHT AND AREA LIMITATIONS

503.1 General. Unless otherwise specifically modified in Chapter 4, building height, number of stories and building area shall not exceed the limits specified in Sections 504 and 506, based on the type of construction as determined by Section 602, and the occupancies as determined by Section 302, except as modified hereafter in this chapter. Building height, number of stories and building area provisions shall be applied independently. Each portion of a building separated by one or more fire walls complying with Section 706 shall be considered to be a separate building.

503.1.1 Special industrial occupancies. Buildings and structures of Type I and II construction designed to house special industrial processes that require large areas and unusual building heights to accommodate craneways or special machinery and equipment, including, among others, rolling mills; structural metal fabrication shops and foundries; or the production and distribution of electric, gas or steam power, shall be exempt from the building height, number of stories and building area limitations specified in Sections 504 and 506. Mezzanines within such buildings or structures shall comply with Section 505.2. Documents and plans describing the special industrial processes shall be submitted to the commissioner for approval.

503.1.2 Buildings on the same tax lot. Two or more buildings on the same tax lot and under the same ownership and control shall be regulated as separate buildings, or, at the option of the owner, shall be considered as portions of one building where the building height, number of stories of each building and the aggregate building area of the combined buildings are within the limitations specified in Sections 504 and 506. The provisions of this code applicable to the aggregate building shall be applicable to each building.

503.1.3 Type I construction. Buildings of Type I construction permitted to be of unlimited tabular building heights and areas are not subject to the special requirements that allow unlimited area buildings in Section 507 or unlimited building height in Sections 503.1.1 and 504.3 or increased building heights and areas for other types of construction.

SECTION BC 504 BUILDING HEIGHT AND NUMBER OF STORIES

504.1 General. The height, in feet, and the number of stories of a building shall be determined based on the type of construction, occupancy classification and whether there is an automatic sprinkler system installed throughout the building.

Exception: The building height of one-story aircraft hangars, aircraft paint hangars and buildings used for the manufacturing of aircraft shall not be limited where the building is provided with an automatic sprinkler system or automatic fire-extinguishing system in accordance with Chapter 9 and is entirely surrounded by public ways or yards not less in width than one and one-half times the building height.

504.1.1 Unlimited area buildings. The height of unlimited area buildings shall be designed in accordance with Section 507.

504.1.2 Special provisions. The special provisions of Section 510 permit the use of special conditions that are exempt from, or modify, the specific requirements of this chapter regarding the allowable heights of buildings based on the occupancy classification and type of construction, provided the special condition complies with the provisions specified in Section 510.

504.2 Mixed occupancy. In a building containing mixed occupancies in accordance with Section 508, no individual occupancy shall exceed the height and number of story limits specified in Section 504 for the applicable occupancies.

504.3 Height in feet and number of stories. The maximum height, in feet, of a building shall not exceed the limits specified in Table 504.3. The maximum number of stories of a building shall not exceed the limits specified in Table 504.4.

Exceptions:

1. Rooftop structures including but not limited to roof tanks and their supports, ventilating, air conditioning, combined heat and power systems and similar building service equipment, bulkheads, penthouses, greenhouses, chimneys, and parapet walls 4 feet (1219 mm) or less in height shall not be included in the building height of the building or considered an additional story unless the aggregate area of all such structures, exclusive of any solar thermal and solar (photovoltaic) collectors and/or panels and their supporting equipment, exceeds $33\frac{1}{3}$ percent of the area of the roof of the building upon which they are erected. Rooftop structures shall be constructed in accordance with Section 1510.
2. Solar thermal and solar electric (photovoltaic) collectors and/or panels and their supporting equipment that exceed $33\frac{1}{3}$ percent of the area of the roof of the building upon which they are erected shall not be included in the height of a building or considered an additional story.
3. Noncombustible steeples, minarets, spires, domes, cupolas and other similar architectural elements, not used for occupancy or storage, shall not be included in the building height of the building or considered an extra story, and may be unlimited in height.

TABLE 504.3^a
ALLOWABLE BUILDING HEIGHT IN FEET ABOVE GRADE PLANE

OCCUPANCY CLASSIFICATION	SEE FOOTNOTES	TYPE OF CONSTRUCTION								
		TYPE I		TYPE II		TYPE III		TYPE IV ^j	TYPE V	
		A	B ⁱ	A	B	A	B	HT	A	B
A-1	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	420	85	75	85	75	85	70	60
A-2	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	420	85	75	85	75	85	70	60
A-3	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	420	85	75	85	75	85	70	60
A-4	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	420	85	75	85	75	85	70	60
A-5	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	420	85	75	85	75	85	70	60
B	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	420	85	75	85	75	85	70	60
E	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	420	85	75	85	75	85	70	60
F-1	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	180	85	75	85	75	85	70	60

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F-2	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	420	85	75	85	75	85	70	60
H-1	S	UL	160	65	55	65	55	65	50	NP
H-2 ^f	S	UL	160	65	55	65	55	65	50	40
H-3 ^f	S	UL	160	65	55	65	55	65	50	40
H-4	S	UL	180	85	75	85	75	85	70	60
H-5	S	UL	160	65	55	65	55	65	50	40
I-1	S13D	60	60	60	NP	60	55	60	NP	NP
	S13R	60	60	60	NP	60	55	60	NP	NP
	S	UL	420	85	NP	85	75	85	NP	NP
I-2	S	UL	180	85	55	65	55	65	50	NP
I-3	S	UL	180	85	75	85	75	85	70	NP
I-4	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	420	85	75	85	75	85	70	60
M	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	420	85	75	85	75	85	70	60
R-1	S13R	60	60	60	NP	60	NP	60	NP	NP
	S	UL	420	85	NP	85	NP	85	NP	NP
R-2	S13R	60	60	60	NP	60	60	60	NP	NP
	S	UL	420	85	NP	85	75	85	NP	NP
R-3	NS ^{b, c}	UL	160	65	55	65	55	65	50 ^{g, h}	40 ^g
	S13D	60	60	60	60	60	55	60	50 ^{g, h}	40 ^g
	S13R	60	60	60	60	60	60	60	60 ^{g, h}	60 ^g
	S	UL	420	85	75	85	75	85	70 ^{g, h}	60 ^g
S-1	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	180	85	75	85	75	85	70	60

(continued)

TABLE 504.3^a—continued
ALLOWABLE BUILDING HEIGHT IN FEET ABOVE GRADE PLANE

OCCUPANCY CLASSIFICATION	SEE FOOTNOTES	TYPE OF CONSTRUCTION								
		TYPE I		TYPE II		TYPE III		TYPE IV ⁱ	TYPE V	
		A	B ^j	A	B	A	B	HT	A	B
S-2 ^{d, e}	NS ^{b, c}	UL	160	65	55	65	55	65	50	40

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	S	UL	420	85	75	85	75	85	70	60
U ^e	NS ^{b, c}	UL	160	65	55	65	55	65	50	40
	S	UL	180	85	75	85	75	85	70	60

 = Not permitted in Fire District

 = Not permitted in Fire District without sprinklers

For SI: 1 foot = 304.8 mm.

Note: UL = Unlimited; NP = Not permitted; NS = Buildings not equipped throughout with an automatic sprinkler system; S = Buildings equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.1; S13R = Buildings equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.2; S13D = Buildings equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.3.

- a. See the following sections for general exceptions to Table 504.3:
 1. See Chapters 4 and 5 for specific exceptions to the allowable height in this chapter.
 2. Section 506.3, Allowable area increase due to frontage.
 3. Section 507, Unlimited area building.
- b. See Section 903.2 for the minimum thresholds for protection by an automatic sprinkler system for specific occupancies.
- c. The NS value is only for use in buildings not required by this code to be equipped throughout with an automatic sprinkler system.
- d. For open parking structures, see Section 406.5.
- e. For private garages, see Section 406.3.
- f. See Section 415.8 for limitations.
- g. See Exception 9 of Section D105.1 for allowances of Type VA and VB construction for R-3 occupancies.
- h. See Exception 12 of Section D105.1 for allowances of Type VA construction and height limitations for R-3 occupancies in Lower Density Growth Management Areas.
- i. See Section 403.2.1 for additional requirements for high-rise buildings.
- j. Type IV construction utilizing cross-laminated timber or structural composite lumber shall be sprinklered where required by Section 903.2.13.

504.4 Number of stories. The maximum number of stories of a building shall not exceed the limits specified in Table 504.4.

TABLE 504.4^{a, b}
ALLOWABLE NUMBER OF STORIES ABOVE GRADE PLANE

[illegible]

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B	NS ^{b, c}	UL	UL	6	3	6	3	6	3	2
	S	UL	UL	7	4	7	4	7	4	3
E	NS ^{b, c}	UL	UL	4	3	4	3	6	3	2
	S	UL	UL	5	4	5	4	7	4	3
F-1	NS ^{b, c}	UL	6	5	3	5	2	5	3	2
	S	UL	7	6	4	6	3	6	4	3
F-2	NS ^{b, c}	UL	UL	6	3	6	3	6	3	2
	S	UL	UL	7	4	7	4	7	4	3

(continued)

TABLE 504.4^{a, b}—continued
ALLOWABLE NUMBER OF STORIES ABOVE GRADE PLANE

OCCUPANCY CLASSIFICATION	SEE FOOTNOTES	TYPE OF CONSTRUCTION								
		TYPE I		TYPE II		TYPE III		TYPE IV ⁱ	TYPE V	
		A	B ⁱ	A	B	A	B	HT	A	B
H-1	S	1	1	1	1	1	1	1	1	NP
H-2 ^f	S	UL	3	2	1	2	1	2	1	1
H-3 ^f	S	UL	6	4	2	4	2	4	2	1
H-4	S	UL	8	6	4	6	4	6	4	3
H-5	S	3	3	3	3	3	3	3	3	2
I-1	S13D	6	6	6	NP	4	3	4	NP	NP
	S13R	UL	UL	6	NP	5	4	5	NP	NP
	S	UL	UL	7	NP	5	4	5	NP	NP
I-2	S	UL	7	5	4	5	3	5	3	NP
I-3	S	UL	5	5	4	5	3	5	4	NP
I-4	NS ^{b, c}	UL	UL	3	3	3	3	3	2	2
	S	UL	UL	4	4	4	4	4	3	3
M	NS ^{b, c}	UL	UL	6	3	6	3	6	3	2
	S	UL	UL	7	4	7	4	7	4	3
R-1	S13R	6	6	6	NP	6	NP	6	NP	NP
	S	UL	UL	7	NP	7	NP	7	NP	NP
R-2	S13R	6	6	6	NP	6	4	6	NP	NP
	S	UL	UL	7	NP	7	4	7	NP	NP
R-3	NS ^{b, c}	✕	✕	✕	✕	✕	✕	✕	3 ^{g, h}	3 ^g

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	S13D	6	6	6	3	6	3	6	3 ^{g, h}	3 ^g
	S13R	6	6	6	4	6	4	6	4 ^{g, h}	4 ^g
	S	UL	UL	7	4	7	4	7	4 ^{g, h}	4 ^g
S-1	NS ^{b, c}	UL	6	5	3	4	3	4	3	2
	S	UL	7	6	4	5	4	5	4	3
S-2 ^{d, e}	NS ^{b, c}	UL	UL	6	3	6	4	6	3	2
	S	UL	UL	7	4	7	5	7	4	3
U ^e	NS ^{b, c}	UL	5	4	2	3	2	4	2	1
	S	UL	6	5	3	4	3	5	3	2

 = Not permitted in Fire District

 = Not permitted in Fire District without sprinklers

 = Number of stories in nonsprinklered 1- & 2-family buildings shall be limited by Section 903.2.8

Note: UL = Unlimited; NP = Not Permitted; NS = Buildings not equipped throughout with an automatic sprinkler system; S = Buildings equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.1; S13R = Buildings equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.2; S13D = Buildings equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.3.

a. See the following sections for general exceptions to Table 504.4.

1. See Chapters 4 and 5 for specific exceptions to the allowable height in this chapter.
2. Section 506.3, Allowable area increase due to frontage.
3. Section 507, Unlimited area building.

b. See Section 903.2 for the minimum thresholds for protection by an automatic sprinkler system for specific occupancies.

c. The NS value is only for use in buildings not required by this code to be equipped throughout with an automatic sprinkler system.

d. For open parking structures, see Section 406.5.

e. For private garages, see Section 406.3.

f. See Section 415.8 for limitations.

g. See Exception 9 of Section D105.1 for allowances of Type VA and VB construction for R-3 occupancies.

h. See Exception 12 of Section D105.1 for allowances of Type VA construction and height limitations for R-3 occupancies in Lower Density Growth Management Areas.

i. Type IV construction utilizing cross-laminated timber or structural composite lumber shall be sprinklered where required by Section 903.2.13.

SECTION BC 505 MEZZANINES AND EQUIPMENT PLATFORMS

505.1 General. Mezzanines shall comply with Section 505.2. Equipment platforms shall comply with Section 505.3.

505.2 Mezzanines. A mezzanine or mezzanines in compliance with Section 505.2 shall be considered a portion of the story below. Such mezzanines shall not contribute to either the building area or number of stories as regulated by Section 503.1. The area of the mezzanine shall be included in determining the fire area. The clear height above and below the mezzanine floor construction shall be not less than 7 feet (2134 mm).

Exception: The clear height of habitable spaces above or below mezzanines within dwelling units shall be not less than 8 feet (2438 mm).

505.2.1 Area limitation. The aggregate area of a mezzanine or mezzanines within a room or space shall be not greater than one-third of the floor area of that room or space in which they are located. The enclosed portions of a room or space shall not be included in determining the permissible floor area of the mezzanine. In determining the

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allowable mezzanine area, the area of the mezzanine shall not be included in the floor area of the room in which it is contained.

Where a room contains both a mezzanine and an equipment platform, the aggregate area of the two raised floor levels shall be not greater than two-thirds of the floor area of that room or space in which they are located.

Exceptions:

1. The aggregate area of mezzanines in buildings and structures of Type I or II construction for special industrial occupancies in accordance with Section 503.1.1 shall be not greater than two-thirds of the area of the room in which the mezzanines are located.
2. The aggregate area of a mezzanine or mezzanines within a dwelling unit shall not exceed one-third of the net floor area of such dwelling unit, whether or not portions of such dwelling unit are enclosed. The area of the mezzanine shall not contribute to the determination of the floor area of the dwelling unit in which it is contained.

505.2.2 Means of egress. Each occupant of a mezzanine shall have access to at least two independent means of egress where the common path of egress travel exceeds the limitations of Section 1006.2. Where a stairway provides a means of exit access from a mezzanine, the maximum travel distance includes the distance traveled on the stairway measured in the plane of the tread nosing. Accessible means of egress shall be provided in accordance with Section 1009.

Exception: A single means of egress shall be permitted in accordance with Section 1006.2.

††† **505.2.3 Openness.** A mezzanine shall be open and unobstructed to the room in which such mezzanine is located except for walls or railings not more than 42 inches (1067 mm) in height, columns and posts.

Exceptions:

1. Mezzanines or portions thereof are not required to be open to the room in which the mezzanines are located, provided that the occupant load of the aggregate area of the enclosed space is not greater than 10.
2. A mezzanine having two or more means of egress is not required to be open to the room in which the mezzanine is located, if at least one of the means of egress provides direct access to an exit located on the mezzanine level.
3. Mezzanines or portions thereof are not required to be open to the room in which the mezzanines are located, provided that the aggregate floor area of the enclosed space is not greater than 10 percent of the mezzanine area.
4. In industrial facilities, mezzanines used for control equipment are permitted to be glazed on all sides.
5. A mezzanine having two or more means of egress shall not be required to be open to the room in which the mezzanine is located in occupancies, other than Groups H and I, that comply with Items 5.1 through 5.4.
 - 5.1. Such occupancy is no more than two stories above grade plane,
 - 5.2. Such occupancy is equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1,
 - 5.3. An approved fire alarm system is installed throughout the entire building or structure in which such occupancy is located, and
 - 5.4. Notification appliances are installed throughout the mezzanine in accordance with NFPA 72.

505.3 Equipment platforms. Equipment platforms in buildings shall not be considered as a portion of the floor below. Such equipment platforms shall not contribute to either the building area or the number of stories as regulated by Section 503.1. The area of the equipment platform shall not be included in determining the fire area in accordance with Section 903. Equipment platforms shall not be a part of any mezzanine and such platforms and the walkways, stairways, alternating tread devices and ladders providing access to an equipment platform shall not serve as a part of the means of egress from the building.

505.3.1 Area limitation. The aggregate area of all industrial equipment platforms within a room shall be not greater than two-thirds of the area of the room in which they occur. Where an equipment platform is located in the same room as a mezzanine, the area of the mezzanine shall be determined by Section 505.2.1 and the combined aggregate

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area of the equipment platforms and mezzanines shall be not greater than two-thirds of the room in which they are located.

505.3.2 Fire suppression. Where located in a building that is required to be protected by an automatic sprinkler system, industrial equipment platforms shall be fully protected by sprinklers above and below the platform, where required by the standards referenced in Section 903.3.

505.3.3 Guards. Equipment platforms shall have guards where required by Section 1015.2.

SECTION BC 506 BUILDING AREA MODIFICATIONS

506.1 General. The floor area of a building shall be determined based on the type of construction, occupancy classification, whether there is an automatic sprinkler system installed throughout the building and the amount of building frontage on public way or open space.

506.1.1 Unlimited area buildings. Unlimited area buildings shall be designed in accordance with Section 507.

506.1.2 Special provisions. The special provisions of Section 510 permit the use of special conditions that are exempt from, or modify, the specific requirements of this chapter regarding the allowable areas of buildings based on the occupancy classification and type of construction, provided the special condition complies with the provisions specified in Section 510.

506.1.3 Basement and cellar. A single basement or cellar need not be included in the total allowable building area, provided such basement or cellar does not exceed the area permitted for a building with no more than one story above grade plane.

506.2 Allowable area determination. The allowable area of a building shall be determined in accordance with the applicable provisions of Sections 506.2.1 through 506.2.4, and Section 506.3.

TABLE 506.2^a
ALLOWABLE AREA FACTOR (A_f = NS, S1, S13D, S13R, or SM, as applicable) IN SQUARE FEET

OCCUPANCY CLASSIFICATION	SEE FOOTNOTES	TYPE OF CONSTRUCTION								
		TYPE I		TYPE II		TYPE III		TYPE IV ⁱⁱ	TYPE V	
		A	B	A	B	A	B	HT	A	B
A-1	NS ^{b, c}	UL	UL	17,500	10,500	14,700	5,600	15,000	8,400	5,500
	S1	UL	UL	52,500	31,500	44,100	16,800	45,000	25,200	16,500
	SM	UL	UL	35,000	21,000	29,400	11,200	30,000	16,800	11,000
A-2	NS ^{b, c}	UL	UL	17,500	9,500	14,000	5,600	15,000	8,400	5,500
	S1	UL	UL	52,500	28,500	42,000	16,800	45,000	25,200	16,500
	SM	UL	UL	35,000	19,000	28,000	11,200	30,000	16,800	11,000
A-3	NS ^{b, c}	UL	UL	17,500	9,500	14,000	5,600	15,000	8,400	5,500
	S1	UL	UL	52,500	28,500	42,000	16,800	45,000	25,200	16,500
	SM	UL	UL	35,000	19,000	28,000	11,200	30,000	16,800	11,000
A-4	NS ^{b, c}	UL	UL	17,500	9,500	14,000	5,600	15,000	8,400	5,500
	S1	UL	UL	52,500	28,500	42,000	16,800	45,000	25,200	16,500

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	SM	UL	UL	35,000	19,000	28,000	11,200	30,000	16,800	11,000
A-5	NS ^{b, c}	UL	UL	UL	UL	UL	UL	UL	UL	UL
	S1	UL	UL	UL	UL	UL	UL	UL	UL	UL
	SM	UL	UL	UL	UL	UL	UL	UL	UL	UL
B	NS ^{b, c}	UL	UL	37,500	10,500	28,500	5,600	36,000	8,400	5,500
	S1	UL	UL	112,500	31,500	85,500	16,800	108,000	25,200	16,500
	SM	UL	UL	75,000	21,000	57,000	11,200	72,000	16,800	11,000
E	NS ^{b, c}	UL	UL	26,000	10,500	23,500	5,600	25,500	8,400	5,500
	S1	UL	UL	78,000	31,500	70,500	16,800	76,500	25,200	16,500
	SM	UL	UL	52,000	21,000	47,000	11,200	51,000	16,800	11,000
F-1	NS ^{b, c}	UL	UL	12,500	7,500	7,500	3,000	10,000	3,000	1,000
	S1	UL	UL	37,500	22,500	22,500	9,000	30,000	9,000	3,000
	SM	UL	UL	25,000	15,000	15,000	6,000	20,000	6,000	2,000

(continued)

TABLE 506.2^a—continued
ALLOWABLE AREA FACTOR (A_t = NS, S1, S13D, S13R, or SM, as applicable) IN SQUARE FEET

OCCUPANCY CLASSIFICATION	SEE FOOTNOTES	TYPE OF CONSTRUCTION								
		TYPE I		TYPE II		TYPE III		TYPE IV ⁱ	TYPE V	
		A	B ⁱ	A	B	A	B	HT	A	B
F-2	NS ^{b, c}	UL	UL	37,500	10,500	28,500	5,600	30,000	8,400	5,500
	S1	UL	UL	112,500	31,500	85,500	16,800	90,000	25,200	16,500
	SM	UL	UL	75,000	21,000	57,000	11,200	60,000	16,800	11,000
H-1	NS ^{b, c}	21,000	16,500	11,000	7,500	9,500	7,000	10,500	7,500	NP
	S1	21,000	16,500	11,000	7,500	9,500	7,000	10,500	7,500	NP
H-2 ^f	NS ^{b, c}	21,000	16,500	11,000	7,500	9,500	7,000	10,500	7,500	3,000
	S1	21,000	16,500	11,000	7,500	9,500	7,000	10,500	7,500	3,000
	SM	21,000	16,500	11,000	7,500	9,500	7,000	10,500	7,500	3,000
H-3 ^f	NS ^{b, c}	UL	60,000	26,500	14,000	17,500	13,000	25,000	10,000	5,000
	S1	UL	60,000	26,500	14,000	17,500	13,000	25,000	10,000	5,000
	SM	UL	60,000	26,500	14,000	17,500	13,000	25,000	10,000	5,000
H-4	NS ^{b, c}	UL	UL	37,000	17,500	28,500	17,500	36,000	18,000	6,500
	S1	UL	UL	112,500	52,500	85,500	52,500	108,000	54,000	19,500

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	SM	UL	UL	75,000	35,000	57,000	35,000	72,000	36,000	13,000
H-5	NS ^{b, c}	UL	UL	37,500	23,000	28,500	19,000	36,000	18,000	9,000
	S1	UL	UL	112,500	69,000	85,500	57,000	108,000	54,000	27,000
	SM	UL	UL	75,000	46,000	57,000	38,000	72,000	36,000	18,000
I-1	NS ^{b, c}	UL	UL	19,000	NP	16,500	5,600	18,000	NP	NP
	S13D	UL	UL	19,000	NP	16,500	5,600	18,000	NP	NP
	S1	UL	UL	57,000	NP	49,500	16,800	54,000	NP	NP
	SM	UL	UL	38,000	NP	33,000	11,200	36,000	NP	NP
I-2	NS ^{b, c}	UL	UL	7,000	3,500	5,000	1,200	6,500	2,000	NP
	S1	UL	UL	21,000	10,500	15,000	3,600	19,500	6,000	NP
	SM	UL	UL	14,000	7,000	10,000	2,400	13,000	4,000	NP
I-3	NS ^{b, c}	UL	UL	7,000	3,500	5,000	1,200	6,500	2,000	NP
	S1	UL	UL	21,000	10,500	15,000	3,600	19,500	6,000	NP
	SM	UL	UL	14,000	7,000	10,000	2,400	13,000	4,000	NP
I-4	NS ^{b, c}	UL	UL	25,500	9,500	23,500	5,600	25,500	8,400	5,500
	S1	UL	UL	79,500	28,500	70,500	16,800	76,500	25,200	16,500
	SM	UL	UL	53,000	19,000	47,000	11,200	51,000	16,800	11,000
M	NS ^{b, c}	UL	UL	21,500	7,500	18,500	5,600	14,000	8,400	5,500
	S1	UL	UL	64,500	22,500	55,500	16,800	42,000	25,200	16,500
	SM	UL	UL	43,000	15,000	37,000	11,200	28,000	16,800	11,000
R-1	NS ^{b, c}	UL	UL	UL	NP	24,000	NP	20,500	NP	NP
	S13R	UL	UL	UL	NP	24,000	NP	20,500	NP	NP
	S1	UL	UL	UL	NP	72,000	NP	61,500	NP	NP
	SM	UL	UL	UL	NP	48,000	NP	41,000	NP	NP
R-2	NS ^{b, c}	UL	UL	UL	NP	24,000	5,600	20,500	NP	NP
	S13R	UL	UL	UL	NP	24,000	5,600	20,500	NP	NP
	S1	UL	UL	UL	NP	72,000	16,800	61,500	NP	NP
	SM	UL	UL	UL	NP	48,000	11,200	41,000	NP	NP

(continued)

TABLE 506.2^a—continued
ALLOWABLE AREA FACTOR (A_t = NS, S1, S13D, S13R, or SM, as applicable) IN SQUARE FEET

OCCUPANCY CLASSIFICATION	SEE FOOTNOTES	TYPE OF CONSTRUCTION								
		TYPE I		TYPE II		TYPE III		TYPE IV ⁱ	TYPE V	
		A	B ⁱ	A	B	A	B	HT	A	B

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R-3	NS ^{b,c}	UL	UL	17,500	10,500	14,700	5,600	30,000	8,400 ^{g,h}	5,500 ^g
	S13D	UL	UL	17,500	10,500	14,700	5,600	30,000	8,400 ^{g,h}	5,500 ^g
	S13R	UL	UL	17,500	10,500	14,700	5,600	30,000	8,400 ^{g,h}	5,500 ^g
	S1	UL	UL	52,500	31,500	44,100	16,800	90,000	8,400 ^{g,h}	5,500 ^g
	SM	UL	UL	35,000	21,000	29,400	11,200	60,000	8,400 ^{g,h}	5,500 ^g
S-1	NS ^{b,c}	UL	48,000	12,000	7,500	7,500	7,500	7,500	5,000	1,000
	S1	UL	144,000	36,000	22,500	22,500	22,500	22,500	15,000	3,000
	SM	UL	96,000	24,000	15,000	15,000	15,000	15,000	10,000	2,000
S-2 ^{d,e}	NS ^{b,c}	UL	UL	15,000	10,000	10,000	8,500	10,000	8,400	5,500
	S1	UL	UL	45,000	30,000	30,000	25,500	30,000	25,200	16,500
	SM	UL	UL	30,000	20,000	20,000	17,000	20,000	16,800	11,000
U ^e	NS ^{b,c}	UL	35,000	19,000	8,500	14,000	8,500	18,000	9,000	5,500
	S1	UL	105,000	57,000	25,500	42,000	25,500	54,000	27,000	16,500
	SM	UL	70,000	38,000	17,000	28,000	17,000	36,000	18,000	11,000

 = Not permitted in Fire District

 = Not permitted in Fire District without sprinklers

Note: UL = Unlimited; NP = Not permitted;

For SI: 1 square foot = 0.0929 m².

NS = Buildings not equipped throughout with an automatic sprinkler system; S1 = Buildings a maximum of one story above grade plane equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.1; SM = Buildings two or more stories above grade plane equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.1; S13R = Buildings equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.2; S13D = Buildings equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.3.

a. See the following sections for general exceptions to Table 506.2.

1. See Chapters 4 and 5 for specific exceptions to the allowable height in this chapter.
2. Section 506.3, Allowable area increase due to frontage.
3. Section 507, Unlimited area building.

b. See Section 903.2 for the minimum thresholds for protection by an automatic sprinkler system for specific occupancies.

c. The NS value is only for use in buildings not required by this code to be equipped throughout with an automatic sprinkler system.

d. For open parking structures, see Section 406.5.

e. For private garages, see Section 406.3.

f. See Section 415.8 for limitations.

g. See Exception 9 of Section D105.1 for allowances of Type VA and VB construction for R-3 occupancies.

h. See Exception 12 of Section D105.1 for allowances of Type VA construction for R-3 occupancies in Lower Density Growth Management Areas.

i. Type IV construction utilizing cross-laminated timber or structural composite lumber shall be sprinklered where required by Section 903.2.13.

506.2.1 Single-occupancy, one-story buildings. The allowable area of a single-occupancy building with no more than one story above grade plane shall be determined in accordance with Equation 5-1:

$$A_a = A_t + (NS \times I_f) \quad \text{(Equation 5-1)}$$

where:

A_a = Allowable area (square feet).

A_t = Tabular allowable area factor (NS, S1, S13D or S13R value, as applicable) in accordance with Table 506.2.

NS = Tabular allowable area factor in accordance with Table 506.2 for nonsprinklered building (regardless of whether the building is sprinklered).

I_f = Area increase factor due to frontage (percent) as calculated in accordance with Section 506.3.

506.2.2 Mixed-occupancy, one-story buildings. The allowable area of a mixed-occupancy building with no more than one story above grade plane shall be determined in accordance with the applicable provisions of Section 508.1 based on Equation 5-1 for each applicable occupancy.

506.2.2.1 Group H-2 or H-3 mixed occupancies. For a building containing Group H-2 or H-3 occupancies, the allowable area shall be determined in accordance with Section 508.4.2, with the sprinkler system increase applicable only to the portions of the building not classified as Group H-2 or H-3.

506.2.3 Single-occupancy, multistory buildings. The allowable area of a single-occupancy building with more than one story above grade plane shall be determined in accordance with Equation 5-2:

$$A_a = [A_t + (NS \times I_f)] \times S_a \quad \text{(Equation 5-2)}$$

where:

A_a = Allowable area (square feet).

A_t = Tabular allowable area factor (NS, S13D, S13R or SM value, as applicable) in accordance with Table 506.2.

NS = Tabular allowable area factor in accordance with Table 506.2 for a nonsprinklered building (regardless of whether the building is sprinklered).

I_f = Area factor increase due to frontage (percent) as calculated in accordance with Section 506.3.

S_a = Actual number of building stories above grade plane, not to exceed three. For buildings equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.2, use the actual number of building stories above grade plane, not to exceed four.

No individual story shall exceed the allowable area (A_a) as determined by Equation 5-2 using the value of $S_a = 1$.

506.2.4 Mixed-occupancy, multistory buildings. Each story of a mixed-occupancy building with more than one story above grade plane shall individually comply with the applicable requirements of Section 508.1. For buildings with more than three stories above grade plane, the total building area shall be such that the aggregate sum of the ratios of the actual area of each story divided by the allowable area of such stories, determined in accordance with Equation 5-3 based on the applicable provisions of Section 508.1, shall not exceed three.

$$A_a = [A_t + (NS \times I_f)] \quad \text{(Equation 5-3)}$$

where:

A_a = Allowable area (square feet).

A_t = Tabular allowable area factor (NS, S13D, S13R or SM value, as applicable) in accordance with Table 506.2.

NS = Tabular allowable area factor in accordance with Table 506.2 for a nonsprinklered building (regardless of whether the building is sprinklered).

I_f = Area factor increase due to frontage (percent) as calculated in accordance with Section 506.3.

Exception: For buildings designed as separated occupancies under Section 508.4 and equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.2, the total building area shall be such that the aggregate sum of the ratios of the actual area of each story divided by the allowable area of such stories determined in accordance with Equation 5-3 based on the applicable provisions of Section 508.1, shall not exceed four.

506.2.4.1 Group H-2 or H-3 mixed occupancies. For a building containing Group H-2 or H-3 occupancies, the allowable area shall be determined in accordance with Section 508.4.2, with the sprinkler system increase applicable only to the portions of the building not classified as Group H-2 or H-3.

506.3 Frontage increase. Area factor increase shall be determined in accordance with Sections 506.3.1 through 506.3.3.

506.3.1 Minimum percentage of perimeter. To qualify for an area factor increase based on frontage, a building shall have not less than 25 percent of its perimeter on a public way or open space. Such open space shall be either on the same zoning lot or dedicated for public use and shall be accessed from a street or approved fire lane.

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506.3.2 Minimum frontage distance. To qualify for an area factor increase based on frontage, the public way or open space adjacent to the building perimeter shall have a minimum distance (W) of 20 feet (6096 mm) measured at right angles from the building face to any of the following:

1. The closest interior tax lot line.
2. The entire width of a street, alley or public way.
3. The exterior face of an adjacent building on the same tax lot.

Where the value of W is greater than 30 feet (9144 mm), a value of 30 feet (9144 mm) shall be used in calculating the building area increase based on frontage, regardless of the actual width of the public way or open space. Where the value of W varies along the perimeter of the building, the calculation performed in accordance with Equation 5-5 shall be based on the weighted average calculated in accordance with Equation 5-4.

$$W = (L_1 \times w_1 + L_2 \times w_2 + L_3 \times w_3 \dots) / F \quad \text{(Equation 5-4)}$$

where:

W = (Width: weighted average) = Calculated width of public way or open space (feet).

L_n = Length of a portion of the exterior perimeter wall.

w_n = Width (≥ 20 feet) of a public way or open space associated with that portion of the exterior perimeter wall.

F = Building perimeter that fronts on a public way or open space having a width of 20 feet (6096 mm) or more.

Exception: Where a building meets the requirements of Section 507, as applicable, except for compliance with the minimum 60-foot (18 288 mm) public way or yard requirement, and the value of W is greater than 30 feet (9144 mm), the value of W shall not exceed 60 feet (18 288 mm).

506.3.3 Amount of increase. The area factor increase based on frontage shall be determined in accordance with Equation 5-5:

$$I_f = (F/P - 0.25)W/30 \quad \text{(Equation 5-5)}$$

where:

I_f = Area increase factor due to frontage.

F = Building perimeter that fronts on a public way or open space having minimum distance of 20 feet (6096 mm).

P = Perimeter of entire building (feet).

W = Width of public way or open space (feet) in accordance with Section 506.3.2.

SECTION BC 507 UNLIMITED AREA BUILDINGS

507.1 General. The area of buildings of the occupancies and configurations specified in Sections 507.1 through 507.18 shall not be limited, except that only one cellar or one basement shall be permissible.

507.1.1 Accessory occupancies. Accessory occupancies shall be permitted in unlimited area buildings in accordance with the provisions of Section 508.2, otherwise the requirements of Sections 507.3 through 507.18 shall be applied, where applicable.

507.2 Measurement of open spaces. Where Sections 507.3 through 507.18 require buildings to be surrounded and adjoined by public ways and yards, those open spaces shall be determined as follows:

1. Yards shall be measured from the building perimeter in all directions to the closest interior lot lines or to the exterior face of an opposing building located on the same tax lot, as applicable.
2. Where the building fronts on a public way, the entire width of the public way shall be used.

507.2.1 Reduced open space. The public ways or yards of 60 feet (18 288 mm) in width required in Sections 507.3, 507.4, 507.5 and 507.6 shall be permitted to be reduced to not less than 40 feet (12 192 mm) in width provided all of the following requirements are met:

1. The reduced width shall not be allowed for more than 75 percent of the perimeter of the building.
2. The exterior walls facing the reduced width shall have a fire-resistance rating of not less than 3 hours.
3. Openings in the exterior walls facing the reduced width shall have opening protectives with a fire protection rating of not less than 3 hours.

507.3 Nonsprinklered, one-story buildings. The area of a Group F-2 or S-2 building no more than one story in height, of other than Type V construction, shall not be limited when the building is surrounded and adjoined on all sides by public ways or yards not less than 60 feet (18 288 mm) in width.

507.4 Sprinklered, one-story buildings. The area of a Group A-4 building no more than one story in height above grade plane of other than Type V construction, or the area of a Group B, F, M or S building no more than one story in height of any construction type, shall not be limited where the building is provided with an automatic sprinkler system throughout in accordance with Section 903.3.1.1 and is surrounded and adjoined on all sides by public ways or yards not less than 60 feet (18 288 mm) in width.

Exceptions:

1. In Group F-2 and S-2 occupancies, one-story rack storage facilities of Type II construction that are not accessible to the public shall not be limited in height provided that such buildings conform to the requirements of the *New York City Fire Code*, and Sections 507.4 and 903.3.1.1 of this code.
2. The automatic sprinkler system shall not be required in areas occupied for indoor participant sports, such as tennis, skating, swimming and equestrian activities, in occupancies in Group A-4, provided that both of the following criteria are met:
 - 2.1. Exit doors directly to the outside are provided for occupants of the participant sports areas.
 - 2.2. The building is equipped with a fire alarm system with manual fire alarm boxes installed in accordance with Section 907.

507.4.1 Mixed occupancy buildings with Groups A-1 and A-2. Group A-1 and A-2 occupancies of other than Type V construction shall be permitted within mixed occupancy buildings of unlimited area complying with Section 507.4, provided all of the following criteria are met:

1. Group A-1 and A-2 occupancies are separated from other occupancies as required for separated occupancies in Section 508.4.4 with no reduction allowed in the fire-resistance rating of the separation based upon the installation of an automatic sprinkler system.
2. Each area of the portions of the building used for Group A-1 or A-2 occupancies shall not exceed the maximum allowable area permitted for such occupancies in Section 503.1.
3. Exit doors from Group A-1 and A-2 occupancies shall discharge directly to the exterior of the building.

507.5 Sprinklered, two-story buildings. The area of a Group B, F, M or S building no more than two stories in height shall not be limited when the building is provided with an automatic sprinkler system in accordance with Section 903.3.1.1 throughout, and is surrounded and adjoined on all sides by public ways or yards not less than 60 feet (18 288 mm) in width.

507.6 Group A buildings. The area of a Group A-1, A-2, A-3, or A-4 building of Type IIA, IIIA or IV construction shall not be limited where the building is equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1. The area of a one-story, Group A-3 building used as a house of worship, community hall, dance hall, exhibition hall, gymnasium, lecture hall, indoor swimming pool or tennis court of Type IIB construction shall not be limited when all of the following criteria are met:

1. The building shall not have a stage other than a platform.
2. The building shall be equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1.
3. The assembly floor shall be located at or within 21 inches (533 mm) of street or grade level and all exits are provided with ramps complying with Section 1012.1 to the street or grade level.

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4. The building shall be surrounded and adjoined on all sides by public ways or yards not less than 60 feet (18 288 mm) in width.

507.7 Reserved.

507.8 Group H occupancies. Group H-2, H-3 and H-4 fire occupancies shall be permitted in unlimited area buildings containing Group F or S occupancies in accordance with Sections 507.4 and 507.5 and the provisions of Sections 507.8.1 through 507.8.4.

507.8.1 Allowable area. The aggregate floor area of Group H occupancies located in an unlimited area building shall not exceed 10 percent of the area of the building and shall not exceed the area limitations for the Group H occupancies as specified in Section 506 based on the perimeter of each Group H floor area that fronts on a public way or open space.

507.8.1.1 Located within the building. The aggregate floor area of Group H occupancies not located at the perimeter of the building shall not exceed 25 percent of the area limitations for the Group H occupancies as specified in Section 506.

507.8.1.1.1 Liquid use, dispensing and mixing rooms. Liquid use, dispensing and mixing rooms having a floor area of not more than 500 square feet (46.5 m²) need not be located on the outer perimeter of the building where they are in accordance with the *New York City Fire Code* and NFPA 30.

507.8.1.1.2 Liquid storage rooms. Liquid storage rooms having a floor area of not more than 1,000 square feet (93 m²) need not be located on the outer perimeter where they are in accordance with the *New York City Fire Code* and NFPA 30.

507.8.1.1.3 Spray paint booths. Spray paint booths that comply with the *New York City Fire Code* need not be located on the outer perimeter.

507.8.2 Located on building perimeter. Except as provided for in Section 507.8.1.1, Group H occupancies shall be located on the perimeter of the building. In Group H-2 and H-3 occupancies, not less than 25 percent of the perimeter of such occupancies shall be an exterior wall.

507.8.3 Occupancy separations. Group H occupancies shall be separated from the remainder of the unlimited area building and from each other in accordance with Table 508.4.

507.8.4 Height limitations. For two-story, unlimited area buildings, Group H occupancies shall not be located more than one story above grade plane unless permitted based on the allowable height and number of stories and feet as set forth in Section 504 based on the type of construction of the unlimited area building.

507.9 Unlimited mixed occupancy buildings with Group H-5. The area of a Group B, F, H-5, M or S building no more than two stories above grade plane shall not be limited where the building is equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1, and is surrounded and adjoined by public ways or yards not less than 60 feet (18 288 mm) in width, provided all of the following criteria are met:

1. Buildings containing Group H-5 occupancy shall be of Type I or II construction.
2. Each area used for Group H-5 occupancy shall be separated from other occupancies as required in Sections 415.11 and 508.4.
3. Each area used for Group H-5 occupancy shall not exceed the maximum allowable area permitted for such occupancies in Section 503.1 including modifications of Section 506.

Exception: Where the Group H-5 occupancy exceeds the maximum allowable area, the Group H-5 shall be subdivided into areas that are separated by 2-hour fire barriers.

507.10 Aircraft paint hangar. The area of a Group H-2 aircraft paint hangar no more than one story above grade plane shall not be limited where such aircraft paint hangar complies with the provisions of Section 412.6 and is surrounded and adjoined by public ways or yards not less in width than one and one-half times the building height.

507.11 Group E buildings. The area of a Group E building of Type IIA, IIIA or IV construction shall not be limited where the building is protected throughout with an approved automatic sprinkler system in accordance with Section 903.3.1.1. The area of a one-story Group E building of Type IIB construction shall not be limited when the following criteria are met:

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1. Each classroom shall have not less than two means of egress, with one of the means of egress being a direct exit to the outside of the building complying with Section 1022.
2. The building is equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1.
3. The building is surrounded and adjoined on all sides by public ways or yards not less than 60 feet (18 288 mm) in width.

507.12 Motion picture theaters. In buildings of Type II, IIIA and IV construction, the area of a motion picture theater located on the first story above grade plane shall not be limited when the building is provided with an automatic sprinkler system throughout in accordance with Section 903.3.1.1 and is surrounded and adjoined on all sides by public ways or yards not less than 60 feet (18 288 mm) in width.

507.13 Covered and open mall buildings and anchor buildings. The area of covered and open mall buildings and anchor buildings not exceeding three stories in height that comply with Section 402 shall not be limited.

507.14 Group B buildings. Notwithstanding the provisions of Sections 507.3 and 507.4, the area of a Group B building of Type IIA, IIIA or IV construction shall not be limited where the building is protected throughout with an approved automatic sprinkler system in accordance with Section 903.3.1.1.

507.15 Group F-2 buildings. Notwithstanding the provisions of Sections 507.3 and 507.4, the area of a Group F-2 building of Type IIA, IIIA or IV construction shall not be limited where the building is protected throughout with an approved automatic sprinkler system in accordance with Section 903.3.1.1.

507.16 Group M buildings. Notwithstanding the provisions of Sections 507.3 and 507.4, the area of a Group M building of Type IIA, IIIA or IV construction shall not be limited where the building is protected throughout with an approved automatic sprinkler system in accordance with Section 903.3.1.1.

507.17 Group R buildings. The area of Group R-1 and R-2 buildings of Type IIA, IIIA or IV construction shall not be limited where the building is protected throughout with an approved automatic sprinkler system in accordance with Section 903.3.1.1 or 903.3.1.2, as applicable.

507.18 Group S-2 buildings. Notwithstanding the provisions of Sections 507.3 and 507.4, the area of a Group S-2 building of Type IB, IIA, IIIA or IV construction shall not be limited where the building is protected throughout with an approved automatic sprinkler system in accordance with Section 903.3.1.1.

SECTION BC 508 MIXED USE AND OCCUPANCY

508.1 General. Each portion of a building shall be individually classified in accordance with Section 302.1. Where a building contains more than one occupancy group, the building or portion thereof shall comply with the applicable provisions of Section 508.2, 508.3 or 508.4, or a combination of these sections.

Exceptions:

1. Occupancies separated in accordance with Section 510.
2. Where required by Table 415.6.2, areas of Group H-1, H-2 and H-3 occupancies shall be located in a detached building or structure.
3. Where identified in Table 509, rooms or spaces shall be treated as incidental uses.
4. Where not identified in Table 509, mechanical and/or electrical equipment rooms shall be permitted to be classified as the occupancy within which they are located, or at the option of the applicant, classified as Group F-2 occupancy in accordance with Section 306.3.
5. Commercial kitchens classified as Group F-2 need not be separated by fire separations from adjoining dining spaces, provided that the conditions of Items 5.1, 5.2, and 5.3 are met:
 - 5.1. The cooking equipment is vented directly to the outdoors;
 - 5.2. A draft curtain of noncombustible material, at least 24 inches (609 mm) down from the ceiling, is provided to separate the opening between the cooking facilities and the dining spaces; and

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5.3. A fire protection system is installed and located as set forth in Item 5.3.1 or 5.3.2.

5.3.1. A fire protection system in accordance with Sections 903 and 904 is installed within the cooking facilities. Additionally, sprinkler heads protecting the opening between the cooking facilities and the dining spaces shall be located within 24 inches (609 mm) of the curtain on the cooking facilities side; or

5.3.2. A fire protection system in accordance with Section 903 is located along any opening between the cooking facilities and the dining space on the cooking facilities side, with sprinkler heads located within 24 inches (609 mm) of the opening and, if the opening is more than 60 inches (1524 mm) wide, the sprinkler heads are spaced not more than 48 inches (1219 mm) on center.

Where the conditions of Items 5.1, 5.2, and 5.3 cannot be satisfied, such commercial kitchen shall be separated from adjoining dining spaces in accordance with Table 508.4, footnote f.

508.2 Accessory occupancies. Accessory occupancies are those occupancies that are ancillary to the main occupancy of the building or portion thereof. Accessory occupancies shall comply with the provisions of Sections 508.2.1 through 508.2.4.

508.2.1 Occupancy classification. Accessory occupancies shall be individually classified in accordance with Section 302.1. The requirements of this code shall apply to each portion of the building based on the occupancy classification of that space.

508.2.2 Allowable building height. The allowable height and number of stories of the building containing accessory occupancies shall be in accordance with Section 504 for the main occupancy of the building.

508.2.3 Allowable building area. The allowable area of the building shall be based on the applicable provisions of Section 506 for the main occupancy of the building. Aggregate accessory occupancies shall not occupy more than 10 percent of the building area of the story in which they are located and shall not exceed the tabular values for nonsprinklered buildings in Table 506.2 for each such accessory occupancy.

Exceptions:

1. The following accessory occupancies are permitted to occupy more than 10 percent of the floor area of the story in which they are located:
 - 1.1. Accessory assembly areas having a floor area less than 750 square feet (69.7 m²).
 - 1.2. Assembly areas that are accessory to Group E occupancies.
 - 1.3. Accessory religious educational rooms and religious auditoriums with occupant loads of less than 100.
2. Rooms or spaces within Group H-2, H-3, H-4 or H-5 occupancy shall not be considered accessory occupancies and shall be treated as separated occupancies.

508.2.4 Separation of occupancies. No separation is required between accessory occupancies and the main occupancy.

Exceptions:

1. Group H-2, H-3, H-4 and H-5 occupancies shall be separated from all other occupancies in accordance with Section 508.4.
2. Separated tenancies in accordance with the requirements of Section 510.

508.3 Nonseparated occupancies. Buildings or portions of buildings that comply with the provisions of this section shall be considered as nonseparated occupancies.

508.3.1 Occupancy classification. Nonseparated occupancies shall be individually classified in accordance with Section 302.1. The requirements of this code shall apply to each portion of the building based on the occupancy classification of that space. In addition, the most restrictive provisions of Chapter 9 that apply to the nonseparated occupancies shall apply to the total nonseparated occupancy area. Where nonseparated occupancies occur in a high-

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rise building, the most restrictive requirements of Section 403 that apply to the nonseparated occupancies shall apply throughout the high-rise building.

508.3.2 Allowable building area and height. The allowable building area and height of the building or portion thereof shall be based on the most restrictive allowances for the occupancy groups under consideration for the type of construction of the building in accordance with Section 503.1.

508.3.3 Separation. No separation is required between nonseparated occupancies.

Exceptions:

1. Group H-2, H-3, H-4 and H-5 occupancies shall be separated from all other occupancies in accordance with Section 508.4.
2. Separated tenancies in accordance with Section 510.10.
3. Commercial kitchens shall be separated in accordance with Section 508.1, Exception 5.

508.4 Separated occupancies. Buildings or portions of buildings that comply with the provisions of this section shall be considered as separated occupancies.

**TABLE 508.4
REQUIRED SEPARATION OF OCCUPANCIES (HOURS)**

OCCUPANCY	A, E		I-1 ^a , I-3, I-4		I-2		R ^a		F-2, S-2 ^b , U		B ^c		F-1		M		S-1		H-1		H-2		H-3, H-4		H-5	
	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS
A, E	N	N	1	2	2	NP	1	2	N ^f	1 ^f	1	2	1	2	1	2	1	2	NP	NP	3	4	2	3	2	NP
I-1 ^a , I-3, I-4	—	—	N	N	2	NP	1	NP	1	2	1	2	1	2	1	2	1	2	NP	NP	3	NP	2	NP	2	NP
I-2	—	—	—	—	N	N	2	NP	2	NP	2	NP	2	NP	2	NP	2	NP	NP	NP	3	NP	2	NP	2	NP
R ^a	—	—	—	—	—	—	N	N	1 ^c	2 ^c	1	2	1	2	1	2	1	2	NP	NP	3	NP	2	NP	2	NP
F-2, S-2 ^b , U	—	—	—	—	—	—	—	—	N	N	1	2	1	2	1	2	1	2	NP	NP	3	4	2	3	2	NP
B ^c	—	—	—	—	—	—	—	—	—	—	N	N	2	3	1	2	2	3	NP	NP	2	3	1	2	1	NP
F-1	—	—	—	—	—	—	—	—	—	—	—	—	N	N	2	3	2	3	NP	NP	2	3	1	2	1	NP
M	—	—	—	—	—	—	—	—	—	—	—	—	—	—	N	N	2	3	NP	NP	2	3	1	2	1	NP
S-1	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	N	N	NP	NP	2	3	1	2	1	NP
H-1	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	N	NP	NP	NP	NP	NP	NP	NP
H-2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	N	NP	1	NP	1	NP
H-3, H-4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	1 ^d	NP	1	NP
H-5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	N	NP

S = Buildings equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.1.

NS = Buildings not equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.1.

N = No separation requirement.

NP = Not permitted.

a. See Section 420.

b. The required separation from areas used only for private or pleasure vehicles shall be reduced by 1 hour but to not less than 1 hour.

c. See Section 406.3.4.

d. Separation is not required between occupancies of the same classification.

e. See Section 422.2 for ambulatory care facilities.

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- f. A commercial kitchen in Group F-2 that does not comply with Section 508.1, Exception 5 shall be separated from adjoining dining spaces with 1-hour fire separation where the dining spaces are equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.1, and separated with 2-hour fire separation where the dining spaces is not equipped throughout with an automatic sprinkler system.

508.4.1 Occupancy classification. Separated occupancies shall be individually classified in accordance with Section 302.1. Each separated space shall comply with this code based on the occupancy classification of that portion of the building.

508.4.2 Allowable building area. In each story, the building area shall be such that the sum of the ratios of the actual building area of each separated occupancy divided by the allowable building area of each separated occupancy shall not exceed 1.

508.4.3 Allowable height. Each separated occupancy shall comply with the building height limitations based on the type of construction of the building in accordance with Section 503.1.

Exception: Special provisions of Section 510 shall permit occupancies at building heights other than provided in Section 503.1.

508.4.4 Separation. Individual occupancies shall be separated from adjacent occupancies in accordance with Table 508.4.

Exception: Fire separations of Group H and I-2 occupancies shall not be permitted any reductions in fire-resistance ratings.

508.4.4.1 Construction. Required separations shall be fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both, so as to completely separate adjacent occupancies.

SECTION BC 509 INCIDENTAL USES

509.1 General. Incidental uses listed in Table 509 and located within single occupancy or mixed occupancy buildings shall comply with the provisions of this section. Incidental uses are ancillary functions associated with a given occupancy that generally pose a greater level of risk to that occupancy and are limited to those uses listed in Table 509.

Exception: Incidental uses within and serving a dwelling unit are not required to comply with this section.

**TABLE 509^c
INCIDENTAL USES**

ROOM OR AREA	SEPARATION AND/OR PROTECTION
Furnace room where any piece of equipment is over 350,000 Btu per hour input	2 hour; or 1 hour and provide automatic sprinkler system ^a
Furnace room where any piece of equipment is 350,000 Btu per hour input or less, except in R-3 occupancy	1 hour or provide automatic sprinkler system ^a
Rooms with a high pressure steam or water boiler that exceeds 350,000 Btu per hour input	2 hour; or 1 hour and provide automatic fire-extinguishing system ^a
Rooms with a high pressure steam or water boiler that is 350,000 Btu per hour input or less	1 hour or provide automatic sprinkler system ^a
Rooms that contain a low pressure steam or water boiler regardless of Btu per hour input	1 hour or provide automatic sprinkler system ^{a, b}
Refrigerant machinery room	1 hour or provide automatic sprinkler system
Hydrogen fuel gas rooms, not classified as Group H	2 hours in all occupancies

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Incinerator rooms	2 hours and provide automatic sprinkler system
Paint shops, not classified as Group H, located in occupancies other than Group F	2 hours; or 1 hour and provide automatic sprinkler system
In Group E occupancies, laboratories and vocational shops not classified as Group H	1 hour or provide automatic sprinkler system
In Group I-2 occupancies, laboratories not classified as Group H	1 hour and provide automatic sprinkler system
In ambulatory care facilities, laboratories not classified as Group H	1 hour or provide automatic sprinkler system
Laundry rooms over 100 square feet	1 hour or provide automatic sprinkler system
In Group I-2, laundry rooms over 100 square feet	1 hour
Group I-3 cells and Group I-2 patient rooms equipped with padded surfaces	1 hour
In Group I-2, physical plant maintenance shops	1 hour
In ambulatory care facilities or Group I-2 occupancies, waste and linen collection rooms with containers that have an aggregate volume of 10 cubic feet or greater	1 hour
In other than ambulatory care facilities and Group I-2 occupancies, waste and linen collection rooms over 100 square feet	1 hour or provide automatic sprinkler system
In ambulatory care facilities or Group I-2 occupancies, storage rooms greater than 100 square feet	1 hour
Stationary storage battery systems having an energy capacity greater than the threshold quantity specified in Table 608.9.1.1 of the <i>New York City Fire Code</i>	1 hour in group B, F, M, S and U occupancies; 2 hours in Group A, E, I and R occupancies
Rooms containing fire pumps in non-high-rise buildings	2 hours; or 1 hour and provide automatic sprinkler system throughout the building
Rooms containing fire pumps in high-rise buildings	2 hours

For SI: 1 square foot = 0.0929 m², 1 pound per square inch (psi) = 6.9 kPa, 1 British thermal unit (Btu) per hour = 0.293 watts, 1 horsepower = 746 watts,

1 gallon = 3.785 L, 1 cubic foot = 0.0283 m³.

- a. Boilers servicing more than one dwelling unit in multiple dwellings shall also comply with Section 65 of the *New York State Multiple Dwelling Law*.
- b. Sealed combustion direct vent boilers shall comply with Section 303 of the *New York City Mechanical Code* and Section 303 of the *New York City Fuel Gas Code*, as applicable.
- c. For mechanical and/or electrical equipment rooms not identified in this table, see Section 508.1.

509.2 Occupancy classification. Incidental uses shall not be individually classified in accordance with Section 302.1. Incidental uses shall be included in the building occupancies within which they are located.

509.3 Area limitations. Incidental uses shall not occupy more than 10 percent of the building in which they are located.

509.4 Separation and protection. The incidental uses listed in Table 509 shall be separated from the remainder of the building or equipped with an automatic sprinkler system, or both, in accordance with the provisions of that table.

509.4.1 Separation. Where Table 509 specifies a fire-resistance-rated separation, the incidental uses shall be separated from the remainder of the building by a fire barrier constructed in accordance with Section 707 or a horizontal assembly constructed in accordance with Section 711, or both. Construction supporting 1-hour fire barriers or horizontal assemblies used for incidental use separations in buildings of Type IIB, IIIB and VB construction is not required to be fire-resistance rated unless required by other sections of this code.

509.4.2 Protection. Where Table 509 permits an automatic sprinkler system without a fire barrier, the incidental uses shall be separated from the remainder of the building by construction capable of resisting the passage of smoke. The walls shall extend from the top of the foundation or floor assembly below to the underside of the ceiling that is a component of a fire-resistance-rated floor assembly or roof assembly above or to the underside of the floor or roof sheathing, deck or slab above. Doors shall be self- or automatic-closing upon detection of smoke in accordance with Section 716.5.9.3. Doors shall not have air transfer openings and shall not be undercut in excess of the clearance permitted in accordance with NFPA 80. Walls surrounding the incidental use shall not have air transfer openings unless provided with smoke dampers in accordance with Section 710.8.

509.4.2.1 Protection limitation. Except as specified in Table 509 for certain incidental uses, where an automatic sprinkler system is provided in accordance with Table 509, only the space occupied by the incidental use need be equipped with such a system.

SECTION BC 510 SPECIAL PROVISIONS

††† **510.1 General.** The provisions in Sections 510.2 through 510.10 shall permit the use of special conditions that are exempt from, or modify, the specific requirements of this chapter regarding the allowable building heights and areas of buildings based on the occupancy classification and type of construction, provided the special condition complies with the provisions specified in this section for such condition and other applicable requirements of this code. The provisions of Sections 510.2 through 510.8 are to be considered independent and separate from each other.

††† **510.2 Horizontal building separation allowance.** Buildings shall be considered as separate and distinct from each other for the purpose of determining area limitations, continuity of fire walls, limitation of number of stories and type of construction, where all of the following conditions are met:

1. The buildings are separated with a horizontal floor assembly having a fire-resistance rating of not less than 3 hours.
2. The building below the horizontal assembly is of Type IA construction.
3. Shaft, stairway, ramp or escalator enclosures through the horizontal floor assembly shall have not less than a 2-hour fire-resistance rating with opening protectives in accordance with Section 716.5.

Exception: Where the enclosure walls below the horizontal floor assembly have not less than a 3-hour fire-resistance rating with opening protectives in accordance with Section 716.5, the enclosure walls extending above the horizontal floor assembly shall be permitted to have a 1-hour fire-resistance rating, provided:

1. The building above the horizontal floor assembly is not required to be of Type I construction;
 2. The enclosure connects fewer than four stories; and
 3. The enclosure opening protectives above the horizontal floor assembly have a fire protection rating of not less than 1 hour.
4. The building or buildings above the horizontal assembly shall be permitted to have multiple Group A occupancy uses, each with an occupant load of less than 300, or Group B, M, R or S occupancies.
 5. The building below the horizontal assembly shall be protected throughout by an approved automatic sprinkler system in accordance with Section 903.3.1.1, and shall be permitted to be any occupancy allowed by this code except Group H.
 6. The maximum building height in feet (mm) as measured from the grade plane shall not exceed the limits set forth in Section 504.3 for the building having the smaller allowable height as measured from the grade plane.

††† **510.3 Group S-2 enclosed parking garage with Group S-2 open parking garage above.** A Group S-2 enclosed parking garage with not more than one story above grade plane and located below a Group S-2 open parking garage shall be classified as a separate and distinct building for the purpose of determining the type of construction where all of the following conditions are met:

1. The allowable area of the building shall be such that the sum of the ratios of the actual area divided by the allowable area for each separate occupancy shall not exceed 1.

2. The Group S-2 enclosed parking garage is of Type I or II construction and is at least equal to the fire-resistance requirements of the Group S-2 open parking garage.
3. The height and the number of the tiers of the Group S-2 open parking garage shall be limited as specified in Table 406.5.4.
4. The floor assembly separating the Group S-2 enclosed parking garage and Group S-2 open parking garage shall be protected as required for the floor assembly of the Group S-2 enclosed parking garage. Openings between the Group S-2 enclosed parking garage and Group S-2 open parking garage, except exit openings, shall not be required to be protected.
5. The Group S-2 enclosed parking garage is used exclusively for the parking or storage of private motor vehicles, but shall be permitted to contain an accessory office, waiting room and toilet room having a total area of not more than 1,000 square feet (93 m²) and mechanical equipment rooms incidental to the operation of the building.

510.4 Parking beneath Group R. Where a maximum one story above grade plane Group S-2 parking garage, enclosed or open, or combination thereof, of Type I construction or open of Type IV construction, with grade entrance, is provided under a building of Group R, the number of stories to be used in determining the minimum type of construction shall be measured from the floor above such a parking area. The horizontal floor assembly between the parking garage and the Group R above shall comply with the type of construction required for the parking garage and shall also provide a fire-resistance rating not less than the mixed occupancy separation required in Section 508.4.

510.5 Reserved.

††† **510.6 Group R-2 buildings of Type IIA construction.** The height limitation for buildings of Type IIA construction in Group R-2 shall be increased to nine stories and 100 feet (30 480 mm) where the building is separated by not less than 50 feet (15 240 mm) from any other building on the lot and from lot lines, the exits are segregated in an area enclosed by a 2-hour fire-resistance-rated fire wall and the first floor assembly has a fire-resistance rating of not less than 1½ hours.

††† **510.7 Open parking garage beneath Groups A, I, B, M and R.** Open parking garages constructed under Groups A, I, B, M and R shall not exceed the height and area limitations permitted under Section 406.5. The height and area of the portion of the building above the open parking garage shall not exceed the limitations in Section 503 for the upper occupancy. The height, in both feet and stories, of the portion of the building above the open parking garage shall be measured from grade plane and shall include both the open parking garage and the portion of the building above the parking garage.

††† **510.7.1 Fire separation.** Fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711 between the parking occupancy and the upper occupancy shall correspond to the required fire-resistance rating prescribed in Table 508.4 for the occupancies involved. The type of construction shall apply to each occupancy individually, except that structural members, including main bracing within the open parking structure, which is necessary to support the upper occupancy, shall be protected with the more restrictive fire-resistance-rated assemblies of the groups involved as shown in Table 601. Means of egress for the upper occupancy shall conform to Chapter 10 and shall be separated from the parking occupancy by fire barriers having not less than a 2-hour fire-resistance rating as required by Section 707, with self-closing doors complying with Section 716 or horizontal assemblies having not less than a 2-hour fire-resistance rating as required by Section 711, with self-closing doors complying with Section 716. Means of egress from the open parking garage shall comply with Section 406.5.

510.8 Industrial uses in buildings containing Group R. No space classified as Factory Industrial Group F shall be located above the second story of any building of Type III, IV or V construction containing a space classified as Residential Group R-1 or R-2.

510.9 Multiple buildings above Group S-2 parking garages. Where two or more buildings are provided above the horizontal assembly separating a Group S-2 open or closed parking garage from the buildings above in accordance with the special provisions in Section 510.2, 510.3 or 510.8, the buildings above the horizontal assembly shall be regarded as separate and distinct buildings from each other and shall comply with all other provisions of this code as applicable to each separate and distinct building.

510.10 Separation of different tenancies. Spaces or dwelling units occupied by different tenants shall be separated by fire barriers having at least 1-hour fire-resistance ratings.

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Exceptions:

1. Nonresidential spaces occupied by different tenants located in buildings that are sprinklered throughout.
2. Tenant spaces in covered mall buildings complying with Section 402.